17 - Group work

(Problem A.4 in *Geometric Approach*.)

Use SAS for the following problem. You can use SAS to avoid hand computations whenever possible.

The heights and sexes of students in a statistics course were recorded as follows:

Sex (M or F):	**F**	**M**	**F**	**M**	**F**	**M**	**F**	**F**	**M**
Height (cm):	140	165	168	184	170	177	156	157	170
Sex:	M	M	M	M	M	M	F	M	
Height:	190	178	187	183	178	175	156	191	

Or in SAS:

```
DATA student_heights;
    INPUT Sex $ Height Female_Dummy;
    DATALINES;
F 140 1
F 168 1
F 170 1
F 156 1
F 157 1
F 156 1
M 165 0
M 184 0
M 177 0
M 170 0
M 190 0
M 178 0
M 187 0
M 183 0
M 178 0
M 175 0
M 191 0
;
RUN;
```

(a) Draw up an observation vector y, with the female heights first and the male heights second.

To enable us to test the hypothesis “females and males are on average equal in height”, we shall pretend that these two samples of female and male heights are random samples from the populations of all adult female heights and all adult male heights. In practice this assumption would require serious examination!! We shall also assume that these two populations are normally distributed with means μ_1 and μ_2 and a common variance σ^2.

(b) Write down the orthogonal decomposition in the form

$$y = \bar{y}_{..} + (\bar{y}_{i\cdot} - \bar{y}_{..}) + (y - \bar{y}_{i\cdot})$$

(Write out each vector.)

(c) Write out the corresponding ANOVA table, and test the hypothesis $H_0 : \mu_1 = \mu_2$. What are your conclusions?